玩转 RISC-V 开发板

从 RuyiSDK 开始

PLCT 实验室
测试团队
25/07/16

关于我

https://github.com/weilinfox/
caiweilin@iscas.ac.cn

蔡玮霖

PLCT 实验室 测试团队
第三测试小队（丙寅）负责人
P119 测试实习生 mentor
RuyiSDK 测试经理

当前主要负责 Ruyi 包管理器、RuyiSDK IDE 等工具的测试工作，ruyisdk.org 的维护工作。同时也参与 RuyiSDK 文档的维护和 Ruyi 软件包的打包。

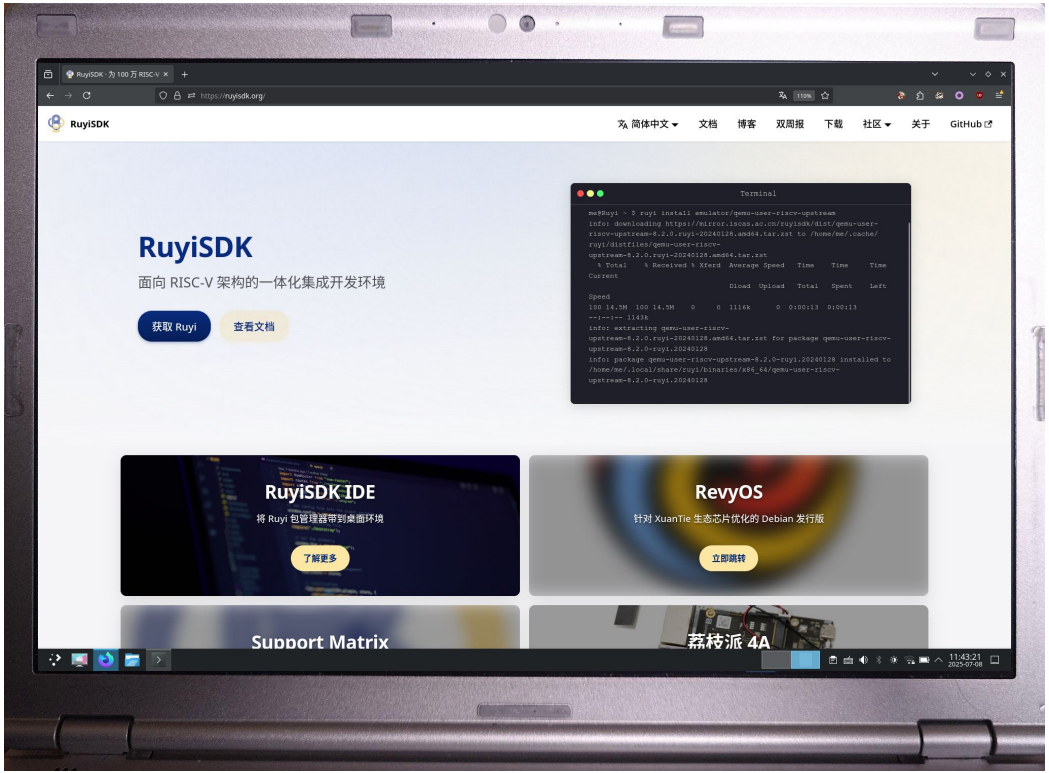
主要内容

- RuyiSDK 简介
- Ruyi 包管理器
- 为 MilkV Meles 刷写最新镜像
- 为 MilkV Duo 刷写最新镜像
- MilkV Duo 开发实例

RuyiSDK 简介

RuyiSDK 是一款专为 RISC-V 开发者打造的全方位、集成式全功能开发环境。RuyiSDK 由中国科学院软件研究所（ISCAS）主导，旨在为 RISC-V 开发者提供一站式的开发解决方案。RuyiSDK 针对玄铁处理器核以及基于处理器核开发的 SOC 发布针对优化的 Linux 发行版（RevyOS）、GCC 工具链、LLVM 工具链，同时发布了 Ruyi 包管理器工具，集成编译环境以及镜像刷写的功能，方便用户充分利用玄铁 RISC-V 处理器核的高性能。

RuyiSDK 简介 ruyisdk.org



- RuyiSDK 文档
- RuyiSDK 生态系统
- RuyiSDK 讨论组(GitHub)
- ruyisdk.cn 开发者社区

Ruyi 包管理器 再也不愁找不到资源

资源同步

Ruyi 包管理器

查询 | 安装 | 更新 | 卸载 | 集成

https://github.com/ruyisdk/ruyi/

存储库（索引及元数据）

https://github.com/ruyisdk/packages-index/

软件源

https://mirror.iscas.ac.cn/ruyisdk/

包括

集成功能

虚拟环境 | 镜像刷写

内容覆盖

索引库和软件源的主要内容

RISC-V 工具链 | RISC-V 模拟器 | 程序及设备调试

RISC-V 分析工具 | RISC-V 设备 SDK | ……

RISC-V 固件及操作系统

运行支持层

操作系统兼容

架构兼容

操作系统兼容性（白底为计划观测）

Debian | Ubuntu | Archlinux | Gentoo | Fedora | Deepin

RevyOS | openEuler | openKylin | OpenCloudOS | ……

操作系统兼容性

x86_64 | riscv64 | aarch64

Ruyi 包管理器 下载和安装

参见官网下载页面

https://ruyisdk.org/download

RevyOS：

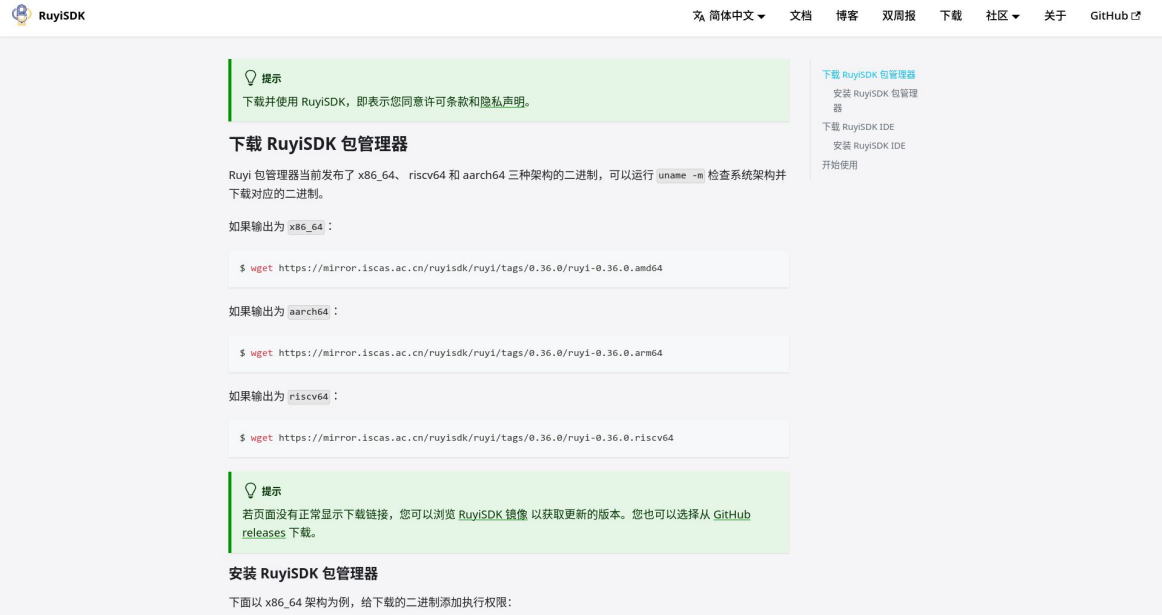
最新镜像都有预装

Gentoo 用户：

https://github.com/ruyisdk/ruyisdk-overlay/

Archlinux 用户添加 Arch Linux CN 源：

sudo pacman -S ruyi



安装后为 Ruyi 包管理器装机数+1

ruyi telemetry upload

命令具体信息参见官网文档

为 MilkV Meles 刷写镜像

Ruyi 镜像刷写功能现场演示：

1. 按住 download 按钮
2. 连接 USB type-c 线
3. ruyi device provision

```
RuyiSDK Device Provisioning Wizard

This is a wizard intended to help you install a system on your device for your
development pleasure, all with ease.

You will be asked some questions that help RuyiSDK understand your device and
your intended configuration, then packages will be downloaded and flashed onto
the device's storage, that you should somehow make available on this host
system beforehand.

Note that, as Ruyi does not run as root, but raw disk access is most likely
required to flash images, you should arrange to allow your user account sudo
access to necessary commands such as dd. Flashing will fail if the sudo
configuration does not allow so.

Continue? (y/N) y

The following devices are currently supported by the wizard. Please pick your device:

  1. Allwinner Nezha D1
  2. BananaPi BPI-F3
  3. Canaan Kendryte K230
  4. Canaan Kendryte K230D
  5. Canaan Kendryte K510
  6. Milk-V Duo
  7. Milk-V Duo S
  8. Milk-V Mars
  9. Milk-V Mars CM
  10. Milk-V Meles
  11. Milk-V Pioneer Box
  12. Milk-V Vega
  13. Pine64 Star64
  14. SiFive HiFive Unmatched
  15. Sipeed Lichee Cluster 4A
  16. Sipeed Lichee Console 4A
  17. Sipeed Lichee RV
  18. Sipeed LicheePi 4A
  19. Sipeed LicheeRV Nano
  20. Sipeed Maix-I
  21. Sipeed Tang Mega 138K Pro
  22. StarFive VisionFive
  23. StarFive VisionFive2
  24. WCH CH32V103 EVB
  25. WCH CH32V203 EVB
  26. WCH CH32V208 EVB
  27. WCH CH32V303 EVB
  28. WCH CH32V305 EVB
  29. WCH CH32V307 EVB
  30. WCH CH582F EVB
  31. WCH CH592X EVB

Choice? (1-31) 10
```

为 MilkV Duo 刷写镜像

Ruyi 镜像刷写功能现场演示：

1. 将 Micro SD 卡连入读卡器
2. 检查设备文件路径
3. ruyi device provision
4. 将 Milkv Duo 连接计算机
5. 检查获得 IP 地址且可以 SSH

```
The following devices are currently supported by the wizard. Please pick your device:

  1. Allwinner Nezha D1
  2. BananaPi BPI-F3
  3. Canaan Kendryte K230
  4. Canaan Kendryte K230D
  5. Canaan Kendryte K510
  6. Milk-V Duo
  7. Milk-V Duo S
  8. Milk-V Mars
  9. Milk-V Mars CM
  10. Milk-V Meles
  11. Milk-V Pioneer Box
  12. Milk-V Vega
  13. Pine64 Star64
  14. SiFive HiFive Unmatched
  15. Sipeed Lichee Cluster 4A
  16. Sipeed Lichee Console 4A
  17. Sipeed Lichee RV
  18. Sipeed LicheePi 4A
  19. Sipeed LicheeRV Nano
  20. Sipeed Maix-I
  21. Sipeed Tang Mega 138K Pro
  22. StarFive VisionFive
  23. StarFive VisionFive2
  24. WCH CH32V103 EVB
  25. WCH CH32V203 EVB
  26. WCH CH32V208 EVB
  27. WCH CH32V303 EVB
  28. WCH CH32V305 EVB
  29. WCH CH32V307 EVB
  30. WCH CH582F EVB
  31. WCH CH592X EVB

Choice? (1-31) 6

The device has the following variants. Please choose the one corresponding to your harc
and:

  1. Milk-V Duo (256M RAM)
  2. Milk-V Duo (64M RAM)

Choice? (1-2) 2

The following system configurations are supported by the device variant you have chose
pick the one you want to put on the device:

  1. Milk-V Duo Official Arduino SDK (64M RAM, SD card)
  2. Milk-V Duo Official buildroot SDK (64M RAM, Lite, SD card)
  3. Milk-V Duo Official buildroot SDK (64M RAM, SD card)

Choice? (1-3)
```

MilkV Duo 开发实例

Ruyi 虚拟环境开发的现场演示：

1. Milkv Duo 已经上电
2. 获取 Ruyi 版本的 envsetup.sh
3. cd i2c/ssd1306_i2c
4. make
5. scp -O ssd1306_i2c \
   root@192.168.42.1:~
6. ssh root@192.168.42.1
7. ./ssd1306_i2c
8. OLED 屏幕有显示

```
[hachi@youmu-sz6 ~]$ mkdir test_duo
[hachi@youmu-sz6 ~]$ cd test_duo/
[hachi@youmu-sz6 test_duo]$ ruyi extract milkv-duo-examples
info: extracting duo-examples-dd39bb35f9d686d2e706407952e964c8dcab8586.tar.gz for package milkv-
duo-examples-0.20241209.0+git.dd39bb3
info: package milkv-duo-examples-0.20241209.0+git.dd39bb3 extracted to current working directory
[hachi@youmu-sz6 test_duo]$ ls
adc    df9gms  envsetup.sh  i2c      libs          pwm        README-zh.md
blink  dht22   hello-world  include  mailbox-test  README.md  spi
[hachi@youmu-sz6 test_duo]$ source envsetup.sh
script_dir: /home/hachi/test_duo
Select Product:
1. Duo (CV1800B)
2. Duo256M (SG2002) or DuoS (SG2000)
Which would you like: 1
CHIP: CV180X
ARCH: riscv64
~/test_duo ~/test_duo
info: skipping already installed package gnu-milkv-milkv-duo-musl-bin-0.20240731.0+git.67688c733
5e7
info: Creating a Ruyi virtual environment at /home/hachi/test_duo/milkv-venv...
info: The virtual environment is now created.

You may activate it by sourcing the appropriate activation script in the
bin directory, and deactivate by invoking `ruyi-deactivate`.

A fresh sysroot/prefix is also provisioned in the virtual environment.
It is available at the following path:

    /home/hachi/test_duo/milkv-venv/sysroot

The virtual environment also comes with ready-made CMake toolchain file
and Meson cross file. Check the virtual environment root for those;
comments in the files contain usage instructions.

~/test_duo
Environment is ready.
«Ruyi milkv-venv» [hachi@youmu-sz6 test_duo]$ cd i2c/ssd1306_i2c/
«Ruyi milkv-venv» [hachi@youmu-sz6 ssd1306_i2c]$ ls
Makefile  ssd1306_i2c.c
«Ruyi milkv-venv» [hachi@youmu-sz6 ssd1306_i2c]$ make
riscv64-unknown-linux-musl-gcc -mcpu=c906fdv -march=rv64imafdcv0p7xthead -mcmodel=medany -mabi=l
p64d -O3 -DNDEBUG -I/home/hachi/test_duo/include/system -o ssd1306_i2c.o -c ssd1306_i2c.c
riscv64-unknown-linux-musl-gcc -mcpu=c906fdv -march=rv64imafdcv0p7xthead -mcmodel=medany -mabi=l
p64d -O3 -DNDEBUG -I/home/hachi/test_duo/include/system -o ssd1306_i2c ssd1306_i2c.o -D_LARGEFIL
E_SOURCE -D_LARGEFILE64_SOURCE -D_FILE_OFFSET_BITS=64 -L/home/hachi/test_duo/libs/system/musl_ri
scv64 -lwiringx
«Ruyi milkv-venv» [hachi@youmu-sz6 ssd1306_i2c]$ file ssd1306_i2c.o
ssd1306_i2c.o: ELF 64-bit LSB relocatable, UCB RISC-V, RVC, double-float ABI, version 1 (SYSV),
not stripped
«Ruyi milkv-venv» [hachi@youmu-sz6 ssd1306_i2c]$ ruyi-deactivate
[hachi@youmu-sz6 ssd1306_i2c]$
```

写在最后

Ruyi 包管理器当前提供超过 75 个软件包，支持超过 31 种开发板。其中包含所有我们可以购买到的玄铁内核的开发板，当然也支持比如 Sifive、StarFive、香山南湖、甚至沁恒微电子的单片机。

这些都是 RuyiSDK 的一部分

RuyiSDK 生态

Ruyi 包管理器

RuyiSDK IDE

操作系统支持矩阵

RevyOS

关注 RuyiSDK 谢谢喵

感谢大家

关注入群
实时获取
项目动态

RuyiSDK 微信公众号

RuyiSDK 微信交流群